

Predictive Modeling and Risk Scoring for Bank Customer Churn

A Machine Learning Approach to Proactive Customer Retention

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Abstract

Customer churn — the loss of existing customers to competitors — represents one of the most significant sources of preventable revenue loss for retail banks. This paper presents an end-to-end machine learning pipeline for predicting bank customer churn and translating model output into an operational risk-scoring framework. Three classification models — Logistic Regression, Random Forest, and XGBoost — were trained on customer demographic, account, and engagement features, with class imbalance addressed through Synthetic Minority Oversampling (SMOTE). The best-performing model was used to generate a continuous 0–100 risk score for each customer, which was then discretized into four actionable risk bands (Low, Medium, High, Critical) paired with recommended retention actions and an estimated value-at-risk metric. Results show that account activity level, geography, number of products held, and age are the strongest predictors of churn, and that a risk-banded approach allows a retention team to prioritize outreach toward the small subset of customers who account for a disproportionate share of at-risk revenue.

1. Introduction

Customer attrition directly erodes a bank's revenue base, and the cost of acquiring a replacement customer is consistently estimated to be several times higher than the cost of retaining an existing one. Traditional churn-management approaches rely on reactive, broad-based retention campaigns applied uniformly across a customer base, which is both inefficient and often too late to change the outcome for a customer who has already decided to leave. A predictive approach instead identifies at-risk customers before they churn, enabling targeted, timely interventions.

This project addresses two connected problems: (1) building an accurate classifier for whether a given customer will churn, and (2) converting that classifier's output into a business-usable risk score that a retention or relationship-management team can act on directly, rather than a raw probability that requires further interpretation.

2. Related Work

Ensemble tree-based methods have become the standard approach for structured/tabular classification problems of this kind. Random Forests (Breiman, 2001) reduce variance by aggregating many decorrelated decision trees, while gradient-boosted tree methods such as XGBoost (Chen & Guestrin, 2016) iteratively fit trees to the residual errors of prior trees, typically achieving strong performance on tabular data with mixed categorical and numeric features. Class imbalance — a near-universal property of churn datasets, since churners are always a minority — is commonly addressed with the Synthetic Minority Oversampling Technique, SMOTE (Chawla et al., 2002), which generates synthetic minority-class examples by interpolating between existing minority samples rather than simply duplicating them.

3. Dataset and Methodology

3.1 Dataset

The dataset consists of 10,000 bank customer records with the following attributes: credit score, geography, gender, age, tenure with the bank, account balance, number of products held, credit card ownership, active-membership status, estimated salary, and a binary churn label (Exited). The overall churn rate in the dataset is approximately 18%, consistent with commonly reported churn rates in retail banking.

3.2 Preprocessing

Categorical variables (geography, gender, and an engineered credit-score band) were one-hot encoded. Numeric features were standardized using z-score scaling fit on the training set only. Data was split 80/20 into training and test sets using stratified sampling to preserve the churn ratio in both partitions.

3.3 Feature Engineering

Three engineered features were added to the base schema to capture relationships not directly present in the raw columns:

- Balance-to-salary ratio — captures financial exposure relative to income
- Zero-balance flag — isolates a distinct customer segment (dormant or transactional-only accounts)
- Credit-score band — discretizes continuous credit score into standard risk tiers (Poor, Fair, Good, Very Good, Excellent)

3.4 Class Imbalance Handling

Because churners represent a minority class, SMOTE oversampling was applied to the training data only, after scaling and before model fitting. This avoids information leakage into the test set while giving the model balanced exposure to churn examples during training.

4. Model Training and Evaluation

Three classifiers were trained and compared under identical preprocessing conditions: Logistic Regression as an interpretable linear baseline, Random Forest as a bagged ensemble capable of capturing non-linear feature interactions, and XGBoost as a gradient-boosted ensemble. Models were evaluated on the held-out test set using accuracy, precision, recall, F1-score, and ROC-AUC, with ROC-AUC used as the primary selection criterion since it is threshold-independent and robust to class imbalance.

Model	Accuracy	Precision	Recall	ROC-AUC
Logistic Regression	0.78	0.41	0.63	0.68
Random Forest	0.81	0.46	0.61	0.69
XGBoost	0.80	0.45	0.62	0.69

All three models achieved comparable ROC-AUC in the 0.68–0.69 range, indicating that churn signal in the dataset is real but moderate in strength — consistent with churn behavior in practice, which is influenced by many factors outside

any single dataset's scope (competitor offers, life events, service incidents, etc.). Random Forest was selected as the production model based on its marginally superior ROC-AUC and precision.

Feature importance analysis identified active-membership status, geography (particularly Germany), number of products held, and age as the strongest predictors of churn, consistent with the patterns surfaced during exploratory data analysis.

5. Risk Scoring Framework

The core practical contribution of this project is converting the model's predicted churn probability into a business-usable risk-scoring system, rather than stopping at classification accuracy. Each customer's predicted probability is scaled to a 0–100 Risk Score and assigned to one of four bands, each paired with a recommended action for the retention team:

Risk Band	Score Range	Customer Count	Recommended Action
Low	0–24	~5,800	Standard monitoring
Medium	25–49	~2,400	Targeted retention campaign
High	50–74	~1,440	Proactive outreach within 7 days
Critical	75–100	~360	Immediate relationship-manager call

An Estimated Value at Risk metric was additionally computed for each customer as the product of account balance and churn probability, giving the retention team a way to prioritize not just by likelihood of churn but by the financial consequence of losing that specific customer. This reframes churn prediction from a pure classification exercise into a resource-allocation tool for the retention team.

6. Results and Discussion

The risk-scoring output concentrated a disproportionate share of estimated revenue exposure into the Critical band: despite representing under 4% of the customer base, Critical-band customers accounted for a substantially higher share of total value at risk than their population share would suggest. This validates the core premise of risk-based prioritization — a small, identifiable subset of customers drives most of the actionable risk, and a retention team with limited capacity should focus there first.

A limitation of this study is that the underlying churn signal, while genuine, has moderate predictive ceiling (ROC-AUC ~0.69), meaning a meaningful share of churn behavior is not explained by the available features. This is expected and typical for churn datasets of this type, where unobserved factors (customer service interactions, competitor promotions, macroeconomic conditions) also drive attrition but are not captured in standard account-level data.

7. Conclusion and Future Work

This project demonstrates a complete pipeline from raw customer data to an operational churn risk-scoring system, showing that model output can be made directly actionable for a non-technical retention team through score banding and value-at-risk estimation. Future work could incorporate: SHAP-based individual-customer explanations to justify each risk score to relationship managers; hyperparameter optimization (e.g., via Bayesian search) to improve model

ROC-AUC; behavioral and transactional time-series features (e.g., recent transaction frequency trends) rather than static account snapshots; and a live-scoring deployment integrated into a bank's CRM system so risk scores update continuously rather than in a single batch run.

References

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